

```
rashiga@localhost:~  
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[rashiga@localhost ~]$ vi sender.c  
[rashiga@localhost ~]$ gcc sender.c -o sender  
[rashiga@localhost ~]$ ./sender  
[rashiga@localhost ~]$
```

```
rashiga@localhost:~  
rashiga@localhost:~  
[rashiga@localhost ~]$ vi receiver.c  
[rashiga@localhost ~]$ gcc receiver.c -o receiver  
[rashiga@localhost ~]$ ./receiver  
Message Received: Welcome to Shared Memory  
[rashiga@localhost ~]$
```

```
rashiga@localhost:~  
[rashiga@localhost ~]$ vi prodcons.c  
[rashiga@localhost ~]$ gcc prodcons.c -o prodcons  
[rashiga@localhost ~]$ ./prodcons  
Produced: 0  
Consumed: 0  
Produced: 1  
Consumed: 1  
Produced: 2  
Consumed: 2  
Produced: 3  
Consumed: 3  
Produced: 4  
Consumed: 4  
Produced: 5  
Consumed: 5  
Produced: 6  
Consumed: 6  
Produced: 7  
Consumed: 7  
Produced: 8  
Consumed: 8  
Produced: 9  
Consumed: 9
```

```
rashiga@localhost:~ -- vi prodcons.c  
#include <stdio.h>  
#include <pthread.h>  
#include <semaphore.h>  
#include <unistd.h>  
#define BUFFER_SIZE 5  
int buffer[BUFFER_SIZE];  
int in = 0, out = 0;  
sem_t mutex, empty, full;  
void* producer(void* arg) {  
    int item;  
    for(int i = 0; i < 10; i++) {  
        item = i;  
        sem_wait(&empty);  
        sem_wait(&mutex);  
        buffer[in] = item;  
        printf("Produced: %d\n", item);  
        in = (in + 1) % BUFFER_SIZE;  
        sem_post(&mutex);  
        sem_post(&full);  
        sleep(1);  
    }  
}
```

```
rashiga@localhost:~ — vi sender.c
rashiga@localhost:~ — vi sender.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#define SharedMemSize 50
void main(){
char c;
int shmid;
key_t key;
char *shared_memory;
key = 5677;
if ((shmid = shmget(key, SharedMemSize, IPC_CREAT | 0666)) < 0){
perror("shmget");
exit(1);
}
if((shared_memory= shmat(shmid, NULL, 0)) == (char *) -1) {
perror("shmat");
exit(1);
}
sprintf(shared_memory," Welcome to Shared Memory");
sleep(2);
exit(0);
}
```

```
rashiga@localhost:~ — vi receiver.c
rashiga@localhost:~ — vi sender.c
rashiga@localhost:~ — vi receiver.c
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#define SharedMemSize 50
void main(){
int shmid;
key_t key;
char *shared_memory;
key = 5677;
if ((shmid = shmget(key, SharedMemSize, 0666)) < 0) {
perror("shmget");
exit(1);
}
if((shared_memory = shmat(shmid, NULL, 0))== (char *) -1) {
perror("shmat");
exit(1);
}
printf("Message Received: %s \n",shared_memory);
exit(0);
-- INSERT --
```

```
void* consumer(void* arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        sem_wait(&full);
        sem_wait(&mutex);
        item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
        sleep(2);
    }
}

int main() {
    pthread_t p, c;
    sem_init(&mutex, 0, 1);
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);
    pthread_join(p, NULL);
    pthread_join(c, NULL);
    -- INSERT --
}
```